Day 02

2 of 30 · Sundays holiday

Skin anatomy & physiology

చర్మ నిర్మాణం & పనితీరు

TODAY'S OBJECTIVES · ఈ రోజు లక్ష్యాలు

- Name the layers of skin and what each layer does
- Understand barrier function, pH and why products sting or burn
- Connect anatomy to treatment depth — why a peel or laser works where it does

THEORY · థియరీ

The three layers

Epidermis (top, ~0.1 mm) — the barrier, renews itself every 28 days; contains melanocytes that make pigment. Dermis (middle) — collagen, elastin, blood vessels, nerves, sweat and oil glands; this is where anti-ageing and scar treatments work. Hypodermis (fat) — cushioning and contour. Treatments are classified by how deep they reach: superficial (epidermis) = peels, Hydrafacial; medium (upper dermis) = MNRF, fractional laser; deep = surgical, HIFU (SMAS).

Barrier, pH and TEWL

The stratum corneum is bricks (dead cells) and mortar (lipids). Healthy skin pH is 4.5–5.5 — slightly acidic. Harsh soap, over-exfoliation or too many actives break the mortar, water escapes (trans-epidermal water loss) and skin becomes sensitive, red and reactive. A damaged barrier must be repaired BEFORE any peel or laser — otherwise you get burns and pigmentation.

Melanin & the sun

Melanocytes sit in the basal layer and pass melanin to skin cells as an umbrella against UV. Indian skin (Fitzpatrick IV–V) has very reactive melanocytes: any inflammation — a pimple, a scratch, an aggressive peel — can trigger post-inflammatory hyperpigmentation (PIH). This single fact decides most of our settings and protocols.

Hands-on today

- Label a skin cross-section diagram from memory (2 attempts)
- Feel and compare barrier quality on 3 volunteers: oily, dry, sensitised
- Map treatment depth on the diagram: peel, Hydrafacial, MNRF, CO2, HIFU

Safety & contraindications

- Never treat a skin with a broken barrier (stinging, redness, flaking) — repair first
- Explain PIH risk to every Indian-skin client before any procedure

KEY POINTS TO REMEMBER · గుర్తుంచుకోవలసినవి

- Epidermis renews ~28 days
- Skin pH 4.5–5.5
- Fitzpatrick IV–V = high PIH risk
- Repair barrier before any active treatment

Homework

Draw the skin cross-section with the 3 layers and mark where a superficial peel, MNRF and HIFU act.

Grooming & professional tip

Clients don't want anatomy — they want to be understood. Learn to say it simply: 'Mee skin barrier weak ga undi, mundu repair chesi tarvata treatment chestam.'

SELF-CHECK · సెల్ఫ్ చెక్

- Did I follow hygiene and consent for every client today?
- Which step did I need help with — and what will I drill tonight?
- Did I write my records and settings before leaving?

Trainer sign: _____ Trainee sign: _____

MY NOTES · నా నోట్స్